

REVIEW ARTICLE

Spinal Epidural Abscess

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SUMMARY

Spinal epidural abscess is an infection in the epidural space. Patients typically present with localized back or neck pain (or both) that is accompanied by fever or neurologic symptoms. Magnetic resonance imaging with contrast enhancement is the diagnostic test of choice. An accurate microbiologic diagnosis is important and can be made with blood cultures, with tissue and fluid cultures obtained by image-guided needle aspiration or biopsy, or at the time of surgery. *Staphylococcus aureus* is the most frequent pathogen, causing more than 50% of infections. All patients with spinal epidural abscess should be promptly evaluated by a spine surgeon and an infectious-disease specialist. Many patients with spinal epidural abscess undergo surgery, although antimicrobial therapy alone may be curative in carefully selected patients.

IN A 1948 ARTICLE IN THE JOURNAL, A. PRICE HEUSNER DESCRIBED HIS 18-year experience treating spinal epidural abscesses at Boston City Hospital.¹ He eloquently detailed the “heavy responsibility accruing to the practitioner who first visits a patient thus afflicted,” and he wrote that the “decisive factor in the outcome of most cases is the celerity with which the first attending physician suspects the probable nature of the ailment and summons expert aid.” Although knowledge of the condition has progressed in the intervening 77 years, diagnostic errors and delays remain common and have important effects on outcomes, and the appropriate approach to management is often unclear. This review will provide an updated understanding of the illness and contemporary approaches to the treatment of patients with spinal epidural abscess.^{2,3}

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CME



CLINICAL FEATURES

A spinal epidural abscess is an infection in the epidural space, located between the dura mater — which surrounds the spinal cord or cauda equina — and the osseous and ligamentous structures of the spine. Local manifestations of spinal epidural abscess have classically been ascribed a sequential, four-phase pattern of symptoms: phase 1, localized pain; phase 2, nerve root pain; phase 3, early neurologic changes (e.g., muscle weakness, sphincter impairment, and sensory deficit); and phase 4, paralysis.¹ Phase 1 symptoms of localized back or neck pain occur in more than 80% of patients.⁴⁻⁷ Although advanced imaging would ideally identify spinal epidural abscess in patients before symptoms reached stage 3 or 4, the diagnosis remains delayed in many patients.³ Stage 3 neurologic symptoms are present at diagnosis in 20 to 47% of patients.^{4,5,8-10} Motor or sensory deficits are reported more commonly than bowel or bladder symptoms.^{4,5,8} Stage 4 symptoms (paralysis or complete neurologic deficit) are present in approximately 3 to 9% of patients.^{5,7,11} The duration of symptoms before diagnosis varies. One third of patients have

symptoms for more than 2 weeks before diagnosis.⁵ Neurologic symptoms are typically present for less than 1 week before diagnosis.⁴ Symptoms of bacterial meningitis may accompany spinal epidural abscess, particularly if a dural tear allows access to the central nervous system. Systemic symptoms of infection are variably present, including fever in 24 to 62% of patients⁴⁻¹⁰ and sepsis in 25 to 35%.^{7,12} Symptoms of nonspecific infection may also occur.¹⁰ Infections at other sites, including endocarditis, are not uncommon.

ANATOMY AND PATHOGENESIS OF INFECTION

The epidural space varies in size and characteristics depending on the vertebral level and whether coexisting illnesses are present, and these factors influence the manifestations and implications of spinal epidural abscess.¹³ The epidural space is commonly divided into ventral (anterior) or dorsal (posterior) spaces but can be considered a circumferential potential space. Within the cervical region, there are minimal fat and veins in the ventral space and no dorsal compartment, because the dura directly contacts the ligamentum flavum and lamina. Beginning at the cervicothoracic junction, the dorsal space progressively enlarges and is greatest at the midlumbar levels. Osteophytes and intervertebral disc protrusions may reduce the size of the ventral epidural space, decreasing the potential volume before a spinal epidural abscess has a neurologic effect.

In general, the potential for neurologic impairment from a spinal epidural abscess within the cervical and thoracic spine is greater than in the lumbosacral spine given the presence of the spinal cord (rather than cauda equina) at those levels, decreased canal volume, and tenuous blood supply in the cervicomedullary and midthoracic regions. Direct compression of the spinal cord or cauda equina is probably the predominant cause of neurologic impairment,¹⁴ although local inflammation leading to thrombosis or vasospasm of the arterial supply of the spinal cord or compression by the abscess on the arterial supply may also occur.¹⁵

The classification of spinal epidural abscesses has varied over the years, limiting the adoption and usefulness of such a system.^{1,10,16,17} The authors prefer to classify spinal epidural abscesses according to the origin, given the implications

for management. Contiguous spinal epidural abscesses are associated with infection of a nearby vertebral or extravertebral structure (Fig. 1). Infected vertebral structures may include intervertebral discs (discitis), vertebral bodies (vertebral osteomyelitis), or facet joints (septic arthritis) or a combination of these. Contiguous extravertebral sites include the paraspinal muscles, retropharyngeal space, posterior mediastinum, posterior pleural space, psoas muscles, and decubitus ulcerations; infections at these sites may occur in conjunction with contiguous vertebral structure infection. Contiguous infection is the most common cause of spinal epidural abscess shown in cohort studies.^{4,5,8} Surgical-site infection—associated spinal epidural abscesses occur after spinal surgery or procedures involving the spine, such as epidural anesthesia, spinal injection procedures, lumbar puncture, or placement of a spinal cord stimulator. Finally, isolated spontaneous spinal epidural abscesses are hypothesized to originate from direct hematogenous seeding of the epidural lymphatic vessels without evidence of contiguous infection.¹⁸ Our opinion is that this type of spinal epidural abscess is rare in the era of widespread use of magnetic resonance imaging (MRI).

The lumbar and lumbosacral levels are the most frequent sites of spinal epidural abscess, accounting for approximately 50% of all infections. Approximately 30 to 40% of spinal epidural abscesses occur in the thoracic spine, with the remaining 10 to 20% involving the cervical spine.^{4,6,8,10,19} Patients with spinal epidural abscess related to injection-drug use are theorized to have more frequent cervical or cervicothoracic infection owing to venous drainage from the arms and hands,²⁰ a finding that has been inconsistently observed.⁸ Most infections span two or three vertebral levels.^{5,6,10} Skip lesions (noncontiguous sites of infection) are commonly observed.²¹ Extensive or holospinal (whole-spine) abscesses that extend from the cervical to lumbosacral vertebrae are uncommon and challenging to manage.²²⁻²⁴ In most studies, ventral spinal epidural abscesses are more common than dorsal spinal epidural abscesses, although circumferential infection can be seen.^{4,5,8,10} The preponderance of ventral spinal epidural abscesses probably reflects the frequency of concomitant vertebral osteomyelitis, spondylodiscitis, and psoas muscle abscesses, collectively found in more than 50% of cases.^{4,5,8}

EPIDEMIOLOGIC AND RISK FACTORS

Spinal epidural abscess remains a relatively rare illness. Historically, the incidence of infection was 0.2 to 1.2 per 10,000 hospital admissions.^{16,25} More-recent studies have shown show a higher incidence of 5.1 to 8.0 infections per 10,000 hospital admissions.^{4,7,12} This increase may relate to wider availability of MRI, but also reflects a true increase in the incidence of the disease, possibly related to increasing rates of spinal surgeries and procedures. Although the epidemic of injection-drug use has contributed to this increase, most patients with spinal epidural abscess do not use injection drugs.^{4-6,8-10,26} Patients with spinal epidural abscess related to injection-drug use are younger and less likely to have diabetes mellitus but have a higher incidence of coexisting liver disease and viral hepatitis than patients with spinal epidural abscess not associated with injection-drug use.^{8,26}

Spinal epidural abscess is primarily a disease of adults (predominantly men) typically between 50 and 65 years of age.^{4-6,9-11,26-29} High-quality case-control studies to define true risk factors are lacking, with one of the only such studies identifying injection-drug use, alcohol abuse, and obesity as independent risk factors.⁴ Other clinical conditions that are frequently observed in cohort studies of patients with spinal epidural abscess³⁰⁻³² are shown in Table 1.

DIAGNOSIS

The classic triad of spinal epidural abscess symptoms — focal pain, fever, and neurologic deficit — is relatively uncommon at the time of diagnosis (present in <20% of patients).^{3,7,33} In contrast, back or neck pain from any cause is nearly ubiquitous among patients seen in acute care settings. When spinal epidural abscess is suspected in a patient whose medical history or clinical features on physical examination reveal the triad of symptoms, the first step in evaluation is to obtain a leukocyte count and inflammatory markers. The erythrocyte sedimentation rate (ESR) and C-reactive protein (CRP) level are inexpensive, rapidly available blood tests that are nonspecific but offer reasonable sensitivity in ruling out the disease. Patients with confirmed spinal epidural abscess typically have markedly elevated levels of ESR and CRP, with commonly reported values

of CRP higher than 100 mg per liter and ESR greater than 60 mm per hour.^{5,7,9,33} An elevated leukocyte count appears to be less sensitive and specific than ESR.³ Which inflammatory marker to use and what the threshold levels should be have been studied but not fully elucidated.^{27,33}

MRI with gadolinium contrast is the diagnostic method of choice to evaluate for spinal epidural abscess.²⁵ Most patients can receive gadolinium-based contrast agents, except in the case of allergy.^{35,36} Intramedullary spinal cord abscess and spinal subdural abscess are rare entities that may complicate spondylodiscitis or occur spontaneously and must be differentiated from spinal epidural abscess. Spinal cord abscess can be differentiated from compressive cord changes with contrast-enhanced MRI to identify a ring-enhancing spinal cord lesion.³⁷ A spinal subdural abscess can be distinguished from a spinal epidural abscess by MRI showing preservation of the epidural space.³⁸ Metallic suppression techniques can be used to decrease (but not eliminate) artifacts from adjacent implants. Patients with cardiac implantable electronic devices often can undergo MRI safely if there is an available support team. An initial MRI to evaluate for an abscess should image the areas of concern on the basis of the location of pain or the level of neurologic deficit. However, once spinal epidural abscess is diagnosed, the remainder of the spine should be imaged with MRI, because noncontiguous abscesses or skip lesions are found in up to 9% of patients.²¹ Among patients who cannot undergo MRI, positron-emission tomography and computed tomography (PET-CT) or CT with intravenous contrast may be an alternative, although both are inferior to MRI, with reported sensitivity of CT as low as 18%.^{39,40} Myelography or CT myelography is invasive and infrequently performed.

Several algorithms have been proposed to identify patients in whom an urgent MRI is warranted, but none have been externally validated or shown to be superior.^{3,27,33,41-43} We propose a modification of previous algorithms to identify patients in whom an urgent MRI is warranted (Fig. 2).^{27,33,34,42,43} For patients who do not undergo an urgent MRI, ongoing evaluation is critical and may lead to MRI at a later time to rule out progressive spinal epidural abscess or other diagnoses.

An accurate microbiologic diagnosis is critical, and all patients with spinal epidural abscess should be evaluated by a spine surgeon and infec-

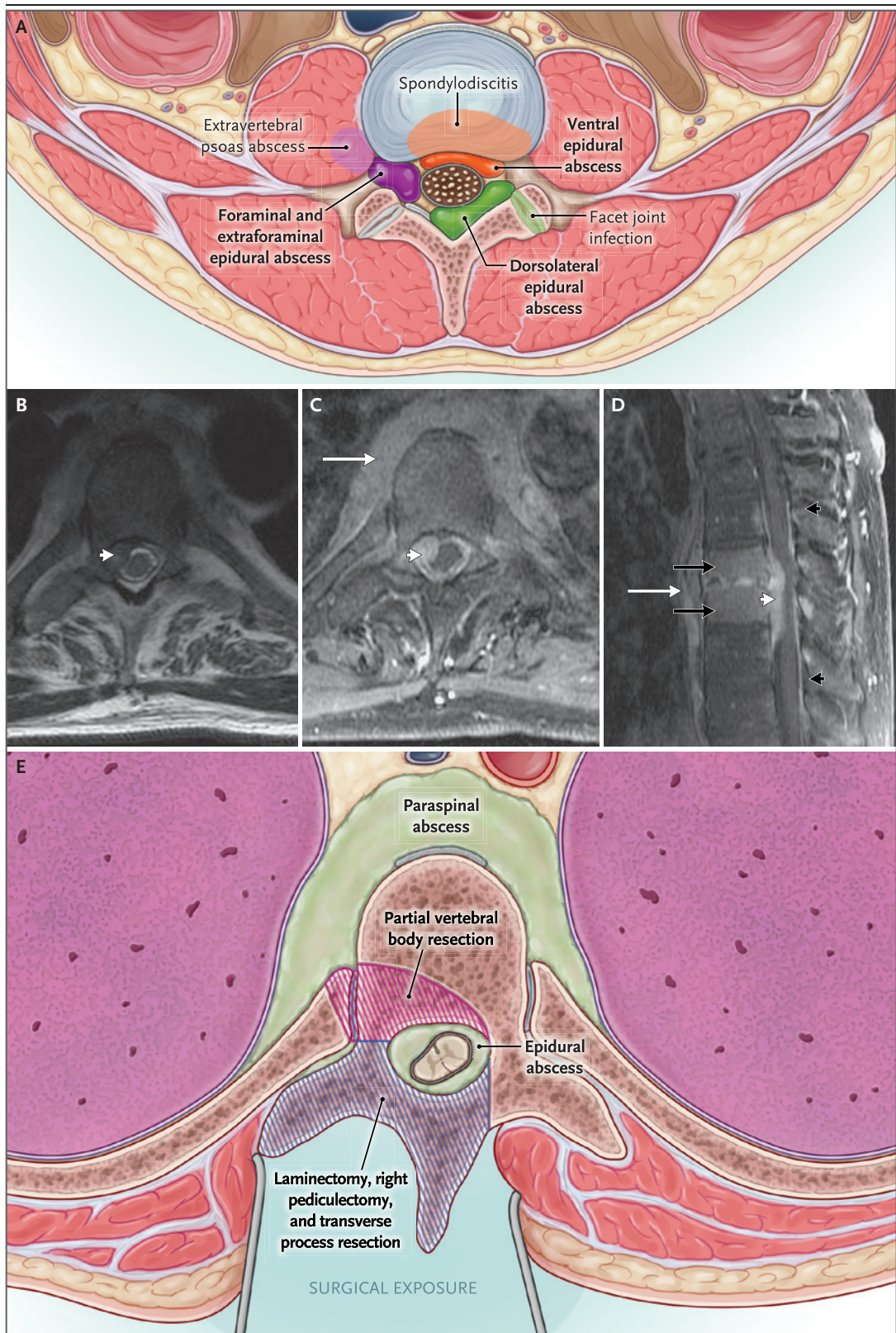


Figure 1 (facing page). Spinal Epidural Abscess and Contiguous Infection of Nearby Structures.

A cross-sectional image of the vertebral structures and surrounding musculature at the level of L3–L4 shows spinal epidural abscesses that are contiguous with vertebral or extravertebral structures (Panel A). In green, a facet joint infection is contiguous with a dorsolateral epidural abscess. In red, spondylodiscitis is contiguous with a ventral epidural abscess. In purple, an extravertebral psoas abscess is contiguous with a foraminal and extraforaminal epidural abscess. A 56-year-old man presented with a 2-week history of thoracic back pain and a 5-day history of progressive weakness and numbness of the legs and feet. MRI (Panel B, axial T2-weighted; Panel C, axial T1-weighted postcontrast with fat saturation at the level of T6–T7; Panel D, sagittal T1-weighted postcontrast with fat saturation) shows enhancement of the T6–T7 vertebral bodies (black arrows) and T6–T7 interspace indicating discitis osteomyelitis. Flagrant enhancing ventral paraspinal abscess (white arrow) displaces the great vessels anteriorly, and right lateral ventrolateral epidural abscess (short white arrows) causes advanced spinal canal stenosis and deformity of the cord. Enhancing ventral and dorsal (short black arrows) epidural phlegmon extends from T5 to T9. The patient underwent emergency epidural abscess evacuation by means of a laminectomy, right facetectomy, and pediclectomy to facilitate transpedicular decompression of the ventral abscess using a posterior approach. Discectomy and partial resection of the vertebral body were performed, followed by posterior fusion using allograft and segmental instrumentation from T5 to T9. Purulent fluid removed from the ventral epidural space during surgery grew *Escherichia coli* and *Bacteroides fragilis*. A posterior surgical approach was used in performance of a laminectomy, right pediclectomy, partial resection of the vertebral body for decompression, and evacuation of an epidural abscess in the thoracic spine (Panel E).

tious-disease specialist. Blood cultures should be obtained before administration of antimicrobial agents. The incidence of bloodstream infection at the time of diagnosis among patients with spinal epidural abscess ranges from 26 to 71%.^{4-6,8,9,17,29} Patients with spinal epidural abscess in whom a microbiologic diagnosis is not made with blood cultures should undergo image-guided biopsy unless surgery is imminent. Interventional neuroradiologists can help decide the safest and highest-yield site to be sampled. Easily accessible contiguous sites of infection, such as psoas muscle abscess, spondylodiscitis, or paraspinal muscle abscess, are preferable to aspiration of epidural fluid in order to avoid inadvertent thecal puncture.^{4,44}

Among patients in whom urgent surgery is planned, image-guided biopsy is unnecessary. At the time of surgery, three to five tissue and fluid samples should be obtained and sent for aerobic and anaerobic bacterial culture with Gram's staining. Blood cultures and surgical or biopsy cultures yield concordant results in approximately 60% of patients.^{4,44,45} Tissue specimens should also be obtained for pathological evaluation to rule out cancer. If available, molecular diagnostics, such as a broad-range bacterial polymerase-chain-reaction assay, should also be performed at the time of surgery.⁴⁵ Liquid specimens of purulent fluid are preferred over swab samples, given the inferior diagnostic accuracy of swab samples.⁴⁵ Cultures should be obtained for tuberculosis, brucellosis, or mycoses endemic to the geographic region, as indicated by the patient's exposure history, along with serologic testing. Finally, cultures for mycobacteria and fungi should be obtained if the patient has immunocompromise.

MICROBIOLOGIC FACTORS

Gram-positive cocci cause most spinal epidural abscesses. *Staphylococcus aureus* is the most frequent pathogen in nearly every study involving patients with a spinal epidural abscess and is responsible for more than 50% of infections (Table 2).⁵⁻¹¹ Other pathogens include the coagulase-negative staphylococci, streptococcus, and enterococcus species. Although gram-negative bacilli are uncommon causes of spinal epidural abscess overall, *Serratia marcescens* is often found in patients with injection-drug use–associated spinal epidural abscess and spondylodiscitis.⁴⁶ Some studies suggest that candida species may be present in as many as 5% of patients with injection-drug use–associated spinal epidural abscess.⁸ In up to 15% of patients, no pathogen is found, a finding that may be attributable to antimicrobial therapy before culture collection.⁵⁻¹⁰

The microbiologic characteristics of spinal epidural abscess may be more nuanced in several circumstances. Contiguous spinal epidural abscess from a retropharyngeal space or an esophageal or other gastrointestinal perforation may have polymicrobial infection that reflects the microbiologic characteristics of the aerodigestive tract. Lumbosacral surgical-site infections have a

Table 1. Clinical Factors Associated with Contiguous Spinal Epidural Abscess and Infection after Spinal Surgery.*

Factors associated with contiguous spinal epidural abscess
Injection-drug use
Diabetes mellitus
End-stage renal disease
Long-term use of indwelling vascular catheters
Immunocompromise conditions
Recent history of endocarditis, bacteremia, or skin or soft-tissue infection
Cirrhosis
Alcohol abuse
Obesity
Recent vertebral fracture
Factors associated with spinal surgical-site infection
Diabetes mellitus
Obesity
Prolonged operative time
Posterior surgical approach
Surgery on seven or more vertebral levels
Hypertension
Intraoperative blood transfusion
Elevated ASA score†
Revision surgery
Tobacco use
Osteotomy
Fusion involving pelvis or sacrum

* There are several clinical factors that are repeatedly identified in patients with contiguous spinal epidural abscesses.^{4,6,9-12,29,33,34} Patients at risk for surgical-site infection (SSI) after spinal surgery would be at increased risk of SSI-associated spinal epidural abscess.³⁰⁻³²

† The American Society of Anesthesiologists (ASA) score is a measure of the physical status of patients undergoing surgery. Scores range from 0 (denoting a healthy patient) to 6 (denoting a patient who has had brain death and whose organs are being removed for donor purposes).

higher proportion of enteric pathogens, including gram-negative bacilli and anaerobes, than infections at the site of cervical or thoracic spinal surgery.⁴⁷ Finally, *Mycobacterium tuberculosis*, brucella species, *Coccidioides immitis*, and *C. posadii* may also cause spinal epidural abscess.

MANAGEMENT

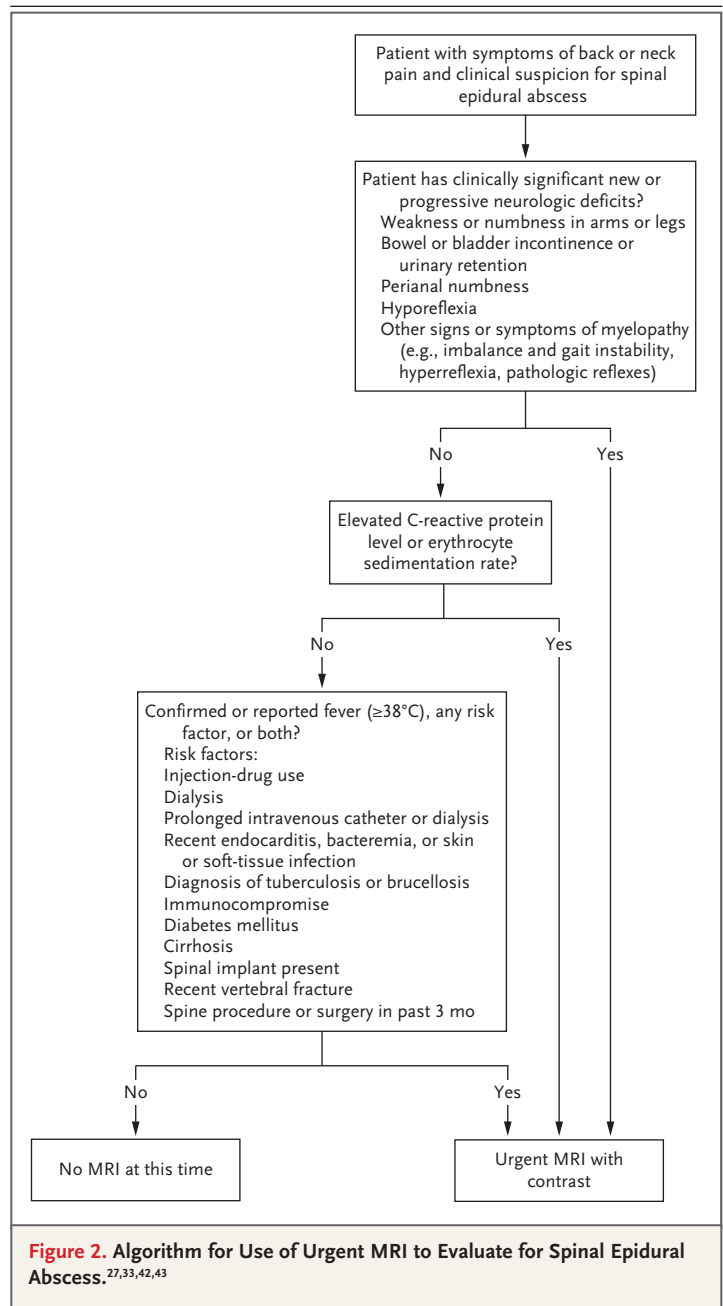
The entire medical team, including the general medical provider, spine surgeon, and infectious-

disease specialist, should work together to restore or preserve neurologic function, cure or control the infection, avoid complications of surgery or antibiotics, and preserve life. Once a diagnosis has been established, time is of the essence, and a decision on whether surgery is indicated or if the patient can be treated nonoperatively is critical. All available tools, including telemedicine visits with an infectious-disease specialist or hospital transfer for spinal surgery, should be used to achieve the best possible outcome for every patient. Historically, all patients with spinal epidural abscess underwent surgery and received antimicrobial therapy. More recently, antimicrobial agents alone or with percutaneous abscess drainage have been used in carefully selected cases.⁴⁸ The optimal strategy has yet to be defined⁴⁹; the approach presented in this review is based on previously published algorithms, studies, and their own practice (Fig. 3).⁵⁰⁻⁵² It should be noted that accumulating medical data may render this approach out of date.

The clearest criterion for surgery is the recent development or progression of neurologic symptoms of spinal epidural abscess. MRI may show cord signal changes on T2-weighted images, a finding that can help determine the urgency of surgery.^{38,53} Cord or cauda equina compression on MRI is typically associated with neurologic deficit at presentation, but may not correlate with outcomes.⁵⁴ Neurologic recovery is most likely among patients who have had symptom onset within the 48 to 72 hours before presentation and the likelihood of recovery decreases progressively with delayed surgery.⁵⁵ In patients with clinically significant new or progressive neurologic deficits, empirical antimicrobial therapy should be immediately initiated after blood-culture collection, and urgent surgery is necessary if they are acceptable candidates.^{20,53,55-57} Attempting nonoperative management in such patients risks permanent neurologic injury that could be avoided with surgery.^{53,55-57} The likelihood of neurologic recovery among patients with spinal cord deficits stemming from spinal epidural abscess appears to be greater than that among patients with traumatic spinal cord injuries.⁵⁸ The exact time that an established neurologic injury becomes permanent is unknown, and some patients recover function even when surgery is delayed.^{56,59} Accordingly, patients with established neurologic injury should be individually evaluated for surgery on the basis of perioperative risk, risk of

worsening neurologic deficits, need for source control, and the likelihood that nonoperative treatment will fail. Other indications for surgery include impending neurologic deficit from spinal instability or progressive deformity. These conditions require both surgical debridement and fusion with instrumentation. Patients who already have instrumentation at the site of a spinal infection are unlikely to be cured by nonoperative treatment, and in most cases, the instrumentation eventually will need to be removed. Finally, surgery is indicated when source control is needed to save the life of the patient.

In every case, the surgical approach must be individualized. Factors to consider include the health of the patient, severity of the neurologic deficit, axial location (ventral, dorsal, or circumferential), vertebral level and extent of the abscess, spinal stability and alignment, and presence of contiguous infection. For example, a patient with a dorsal spinal epidural abscess without vertebral osteomyelitis may be a candidate for a simple decompression with laminectomy, under the assumption that stability can be preserved, whereas a patient with vertebral osteomyelitis and ventral abscess may undergo a corpectomy followed by an instrumented fusion, performed either anteriorly or posteriorly (depending on the region of the spine) and in one or multiple stages (Fig. 1B and 1C). Spinal epidural abscesses are dynamic processes, and the initial plan may need to be changed on the basis of intraoperative findings. Additional vertebral levels may need to be decompressed if more compression is encountered than was anticipated on the basis of findings on MRI, and a fusion may be indicated if the spine is rendered unstable by the infection or the decompression. The material causing the abscess may be partially or totally composed of granulation tissue rather than drainable pus, and a different approach may be needed as a result. When instrumentation is in use, performance of the initial debridement and the definitive fusion in separate stages may be desirable in order to minimize contamination of the instrumentation, although a one-stage approach may be reasonable with appropriate antimicrobial therapy.⁶⁰ Contiguous infection from an extravertebral site, such as a retropharyngeal abscess or esophageal perforation, necessitates definitive management of the primary source of the infection. Multi-level laminectomy may be needed if the abscess is extensive. Alternatively, some studies describe



the use of focused laminectomies at the area or areas where the abscesses will have the greatest mass effect or incorporation of catheter-based irrigation to avoid extensive laminectomy.^{22,24}

Among patients without an indication for urgent surgery, the goal is to identify patients who can be successfully treated nonoperatively. Nonoperative treatment fails in approximately 28% of patients and in those cases leads to delayed surgery.⁵⁷ Delayed surgery has worse outcomes than early surgery.^{49,56,59,61} Multiple studies

Table 2. Pathogens Associated with Spinal Epidural Abscess.*

Pathogen	Infections Attributable to Pathogen or Pathogen Group
	%
Methicillin-susceptible <i>Staphylococcus aureus</i>	40
Methicillin-resistant <i>S. aureus</i>	18
Coagulase-negative staphylococcus species	5
Streptococcus or enterococcus species	10
Aerobic gram-negative bacilli	4
Other pathogens	6
Polymicrobial infections	2
No growth or unknown	14

* Shown are data from 2282 patients included in observational studies.⁵⁻¹¹

have shown that factors associated with failure of nonoperative treatment include advanced age, diabetes mellitus, bloodstream infection, infection with methicillin-resistant *S. aureus* (MRSA), markedly elevated CRP levels, serious neurologic deficit on presentation, leukocytosis, and active cancer.^{6,9,11,62-65} Among nonoperative predictive models using multivariate analysis^{6,11,62,63} or machine-learning techniques,^{64,66} external evaluation suggests that the model developed by Kim et al. has good discriminative ability to predict failure.⁶⁷ In that study, an age greater than 65 years, diabetes mellitus, MRSA infection, and partial or complete spinal cord deficit were independently associated with an increased risk of failure.¹¹

In carefully selected patients, therapeutic image-guided aspiration of a spinal epidural abscess may be used with nonoperative management to decrease the bacterial burden, and this approach has the potential to improve outcomes.^{23,52,68} Potential candidates include patients with a dorsal or dorsolateral abscess or abscesses that appear to contain fluid on MRI. However, this approach requires an experienced interventional radiologist, and there is insufficient data on the safety and effectiveness in large numbers of patients to routinely recommend its use.

All patients with spinal epidural abscess need prolonged antimicrobial therapy guided by an accurate microbiologic diagnosis. Among patients who are neurologically intact without sepsis or spinal cord compression on MRI and who can be carefully monitored for neurologic symptoms,

antibiotics should be withheld until an image-guided biopsy can be performed to enhance diagnostic yield. Patients who do not meet these criteria should receive empirical antimicrobial agents immediately after a blood culture is collected. Parenteral antimicrobial therapy is most often used first; vancomycin with ceftriaxone is one appropriate empirical-therapy approach in patients who are not allergic to the agents and for whom the side-effect profile is acceptable. Cefepime should replace ceftriaxone in patients who use injection drugs or who have a spinal epidural abscess resulting from a surgical-site infection or have sepsis or substantial neurologic deficits. Empirical therapy should be guided by consultation with an infectious-disease specialist as soon as possible. Therapy should be rationalized on the basis of the pathogen and susceptibility testing, with guidelines for management of vertebral osteomyelitis presenting relevant options for pathogen-directed therapy.^{69,70} Unless the patient has clinical meningitis, treatment that penetrates the central nervous system is unnecessary.

Although direct clinician experience in the treatment of spinal epidural abscess is limited, data from other serious bone and joint infections suggest that patients in whom there is an adequate initial response to therapy and a susceptible organism may be treated with the use of highly bioavailable oral antimicrobial agents.⁷¹ Among the 10 to 20% of patients without an identified pathogen, administration of antimicrobial treatment before culture collection is a common cause of negative cultures; in such cases, previously administered antimicrobial agents should be incorporated into subsequent therapy. For example, a patient with negative cultures after exposure to cefazolin should receive definitive treatment that covers pathogens treated by cefazolin. A patient with negative cultures who had not received antimicrobials before culture collection should be evaluated for pathogens with special identification requirements, such as mycobacteria, fungi, or brucella species, according to the risk of those illnesses in that patient.

Throughout therapy, neurologic status must be followed closely. Patients should be counseled to report any changes in neurologic symptoms and to monitor their level of pain. Modifiable risk factors and coexisting conditions must be

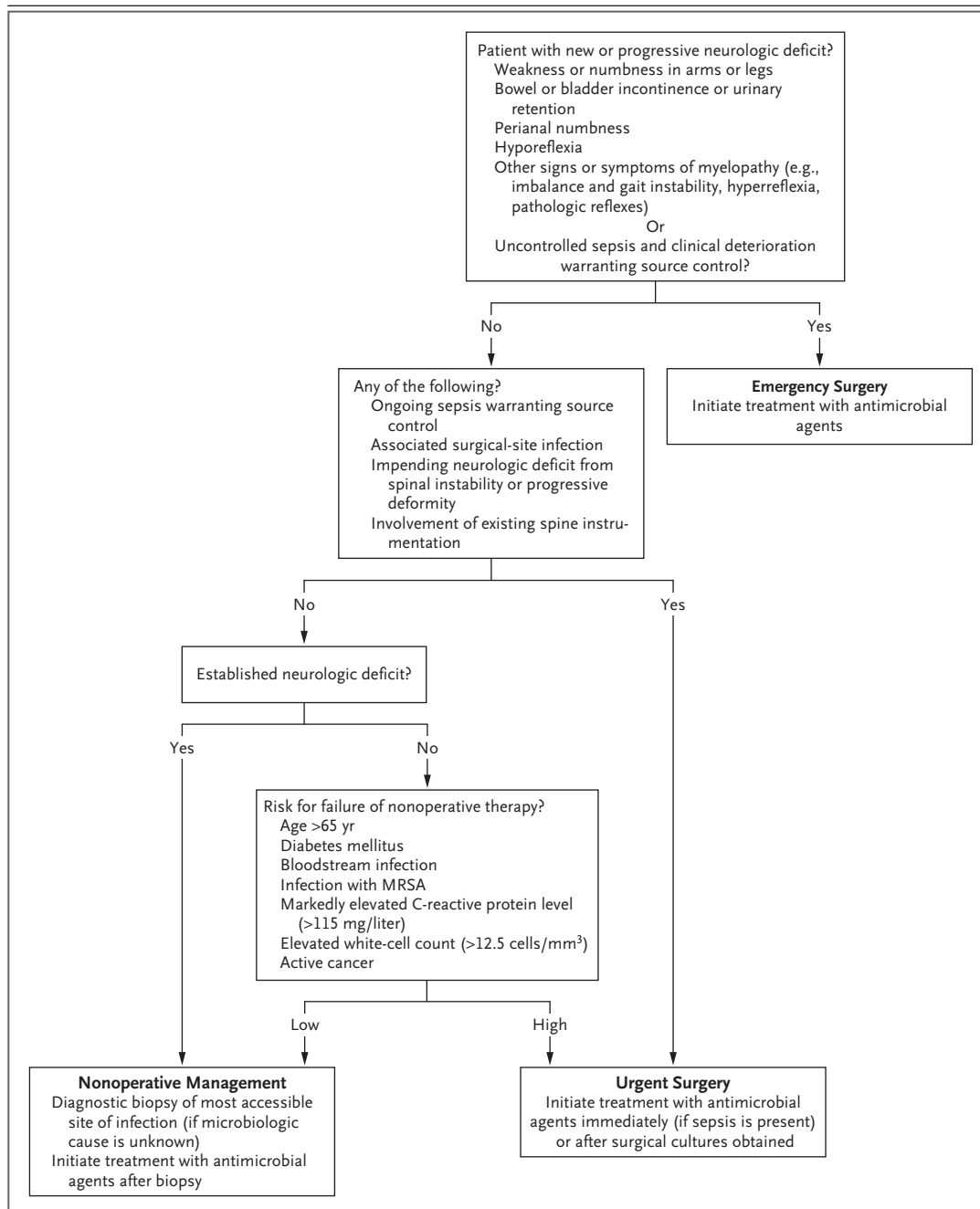


Figure 3. Initial Decision Regarding Surgical versus Nonoperative Treatment.

The duration beyond which recovery from a neurologic deficit is not possible is unknown, but outcomes are best when surgery occurs within 72 hours. However, this estimate is not a hard cutoff time; the decision must be determined collaboratively by the surgeon, infectious-disease specialist, and the medical team, and the treatment should be individualized on the basis of the patient's perioperative risk, the need for source control, the risk of worsening neurologic deficits, and the likelihood of failure of nonoperative treatment. Presumably, the best outcome would occur when surgery is undertaken at the earliest point in time that both the patient and surgical team are adequately prepared. Antimicrobial agents should be pathogen directed when the microbiologic profile is known. Otherwise, an empirical antimicrobial agent should be chosen in collaboration with the infectious-disease specialist. CRP denotes C-reactive protein, and MRSA methicillin-resistant *Staphylococcus aureus*.

KEY POINTS

SPINAL EPIDURAL ABSCESS

- Spinal epidural abscess is an infection in the epidural space; patients typically present with localized back or neck pain (or both) that is accompanied by fever or neurologic symptoms.
- Concomitant infection involving intervertebral discs (discitis), vertebral bodies (vertebral osteomyelitis), or facet joints (septic arthritis) — or a combination of these — is common.
- Magnetic resonance imaging with contrast enhancement is the diagnostic test of choice.
- An accurate microbiologic diagnosis is important and can be made with blood cultures or with tissue and fluid cultures obtained by image-guided needle aspiration or biopsy or at the time of surgery. *Staphylococcus aureus* is the most frequent pathogen, causing more than 50% of infections.
- All patients with spinal epidural abscess should be promptly evaluated by a spine surgeon and an infectious-disease specialist to determine the appropriate medical and surgical treatment to achieve the best outcome.
- Although many patients with spinal epidural abscess undergo surgery, antimicrobial therapy alone may be curative in carefully selected patients.

addressed regardless of the treatment plan. The duration of antimicrobial treatment for spinal epidural abscess is similar to that for native vertebral osteomyelitis (a condition that frequently coexists with spinal epidural abscess).^{69,70} There are insufficient data with which to determine the appropriate duration of therapy, with most patients initially treated for 6 to 8 weeks and additional therapy individualized to the clinical response. The neurologic examination and degree of pain should be evaluated at the proposed completion of antimicrobial therapy; the authors obtain repeat measurements of ESR and CRP levels to compare with baseline levels. Repeat MRI does not appear to be superior to clinical evaluation and monitoring of ESR and CRP levels, as evidenced by data from patients with native vertebral osteomyelitis with or without epidural abscess.⁷² However, among patients with worsening clinical status or increasing or unchanging ESR and CRP levels, MRI can provide information to guide treatment. Changes in the soft tissue on subsequent MRI are the most important indicator of clinical response. Although the authors do not routinely obtain an end-of-therapy MRI, we do so in patients who are nonoperatively treated, have risk factors for failure (e.g., MRSA infection), or do not have abscess drainage. When instrumented fusion is used at the site of a spinal epidural abscess, the authors favor suppressive antimicrobial therapy until bony fusion is shown to be solid on CT and flexion–extension plain radiography. Once the fusion has completely healed, if evidence suggests persistent infection, the instrumentation can be removed

and a thorough irrigation and debridement of the site performed. This procedure usually definitively resolves the infection.

OUTCOMES

There is no standard structure for reporting treatment outcomes in spinal epidural abscess. Approximately 3 to 6% of patients with the disease die during hospitalization,^{12,28,61,73} and 6 to 13% die within 90 days after hospitalization.^{29,52,74-76} Although some recent data suggest lower mortality with surgery than with nonoperative management,²⁹ there are conflicting data from other high-quality studies with regard to this finding.⁷⁵ Among the patients who survive, most are cured, and recurrence is uncommon.⁷⁷ Neurologic outcomes are most accurately predicted by the patient's neurologic status at the time of surgery or initiation of therapy. One meta-analysis showed a good neurologic outcome in 86% of patients with early surgical intervention as compared with 69% with nonoperative treatment.⁴⁹ However, these data may not fully reflect the experience of patients selected for nonoperative treatment, which highlights the importance of individualized treatment decisions.

CONCLUSIONS AND FUTURE DIRECTIONS

Considerable challenges remain with regard to the understanding, diagnosis, and management of spinal epidural abscess. Close cooperation between the medical team, infectious-disease specialist,

and the surgeon are essential to achieving the best possible outcome. Future studies should focus on the evaluation and refinement of published diagnostic algorithms to hasten diagnosis. The explosion of knowledge about the disease and the availability of advanced statistical models have the potential to improve the identification of patients who can be successfully treated nonoperatively. Therapeutic image-guided aspiration may be a valuable adjunct to nonoperative management, but this technically challenging procedure deserves further investigation before routine implementation. Finally, establishment of standardized classification, outcome defini-

tions, and reporting would better allow the study of this uncommon but important illness in a systematic manner, providing clear answers with regard to the appropriate management approach.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

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